As already mentioned, there are two possible cases of unsuccessful search in tries: in the first one, shown in figure 6.6, we traverse the trie until we get to a node that doesn't have an outgoing edge for the next character: in the example, a call to `trie.search("any")`, this happens when we get to the key's last character, `'y'` (as shown in the rightmost diagram in figure 6.6). In listing 6.2, this corresponds to the condition in the `if` at line #5 returning `true`.

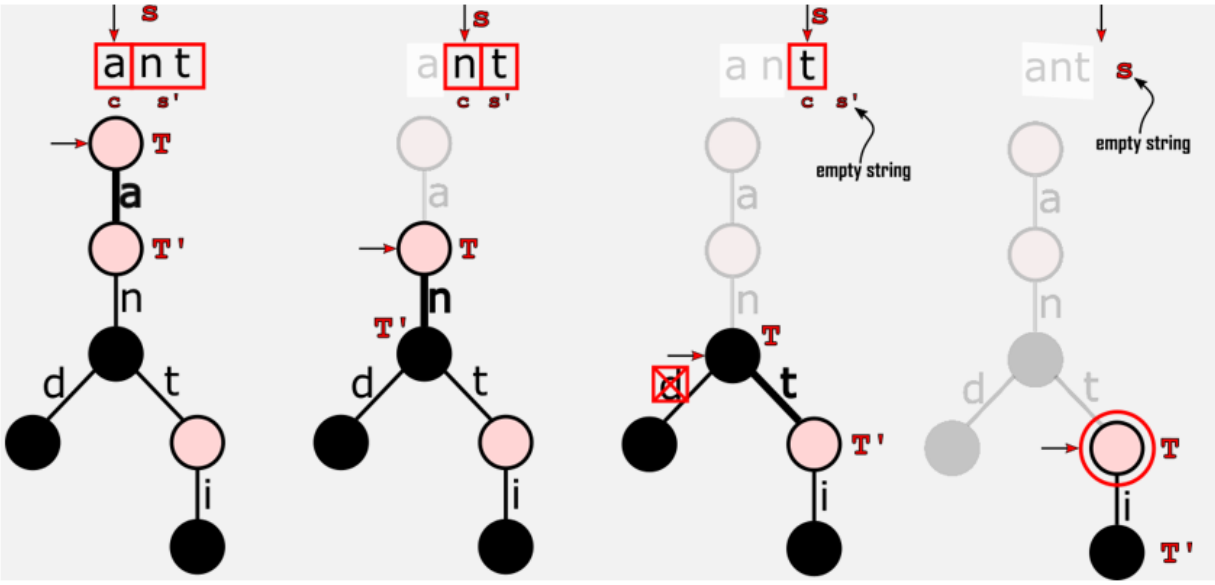


Figure 6.7 Another example of an unsuccessful search in a trie. In this example, the search fails because the path corresponding to the string searched ends in an *intermediate node*.

The other possible case for unsuccessful search is that we always find a suitable outgoing edge, until we recursively call search on the empty string: when this happens, it means we reached the trie's node whose path from the root spells out the searched key; for instance, in figure 6.7 we traverse the